

B.Tech Degree VI Semester Examination April 2012

ME 603 MACHINE DESIGN I (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A

(Answer ALL questions)

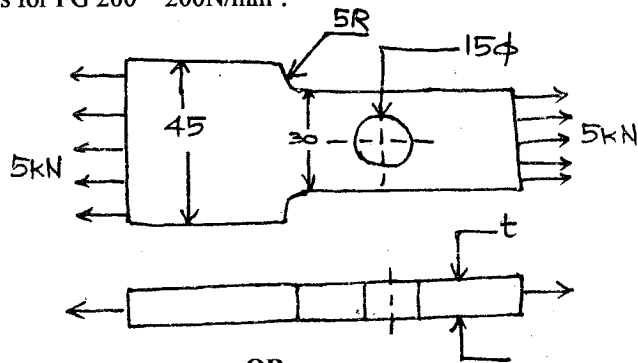
(8 x 5 = 40)

- I. (a) Write short notes on:
- (i) Interchangeability (ii) Preferred Numbers
 - (b) What is meant by stress concentration? With the help of suitable diagrams, explain any two methods of reducing stress concentration.
 - (c) What are bolts of uniform strength? Give examples of practical applications of such bolts.
 - (d) Distinguish between cotter joint and knuckle joint.
 - (e) Write short notes on: (i) Lozenge joint (ii) Caulking and fullering.
 - (f) What is nipping in a leaf spring? Discuss its role.
 - (g) What are the main advantages and disadvantages of welded joints?
 - (h) Under what circumstances are hollow shafts preferred over solid shafts? Give any two examples where hollow shafts are used.

PART B

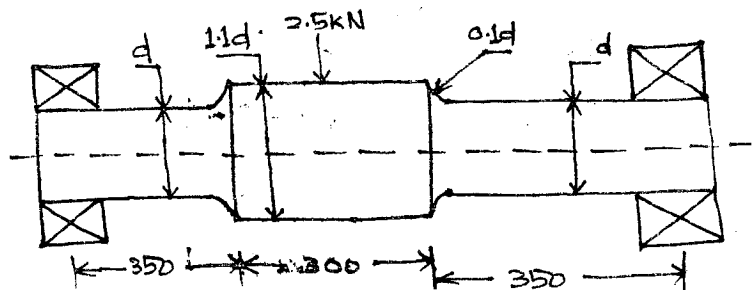
(4 x 15 = 60)

- II. (a) In an elastic material, principal stresses are tensile and compressive and the ratio being 4:1. Determine the limiting stress according to different theories of failure if tension test gives elastic limit of the material as 400N/mm^2 . Assume Poisson's ratio as 0.3. (7)
- (b) A flat plate subjected to a tensile force of 5kN as shown in figure. The plate material is grey cast iron FG 200 and the factor of safety is 2.5. Determine the thickness of plate. (8)
- Ultimate stress for FG 200 = 200N/mm^2 .



OR

- III. A non-rotating shaft supporting a load of 2.5kN is shown in figure. The shaft is made of brittle material with an ultimate tensile strength of 300N/mm^2 . The factor of safety is 3. Determine the dimensions of the shaft. (15)



(P.T.O)

- IV. A vertical two start square threaded screw of a 100mm mean diameter and 20mm pitch supports a vertical load of 18 kN. The axial thrust on the screw is taken by a collar bearing of 250mm outside diameter and 100mm inside diameter. Find the force required at the end of a lever which is 400mm long in order to lift and lower the load. The coefficient of friction for the vertical screw and nut is 0.15 and that for collar bearing is 0.20. (15)

OR

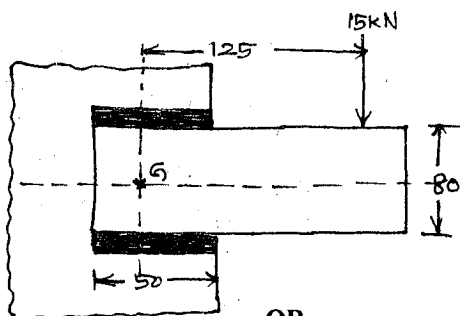
- V. Design and draw a cotter joint to support a load varying from 30 kN in compression to 30kN in tension. The material used is carbon steel for which the following allowable stresses may be used. The load is applied statically. (15)
Tensile stress = Compressive stress = 50MPa;
Shear stress = 35 MPa and crushing stress = 90 MPa

- VI. Design a double rivetted butt joint with two cover plates for the longitudinal seam of boiler shell 1.5m in diameter subjected to a steam pressure of 0.95 N/mm². Assume joint efficiency as 75%, allowable tensile stress in the plate 90MPa; compressive stress 140MPa and shear stress in the rivet 56 MPa. (15)

OR

- VII. A semi-elliptic leaf spring used for automobile suspension consists of three extra full-length leaves and 15 graduated-length leaves, including the master leaf. The centre-to-centre distance between two eyes of the spring is 1m. The maximum force that can act on the spring is 75 kN. For each leaf, the ratio of width to thickness is 9:1. The modulus of elasticity of the leaf material is 207000 N/mm². The leaves are pre-stressed in such a way that when the force is maximum, the stresses induced in all leaves are same and equal to 450 N/MM². Determine: (15)
(i) the width and thickness of the leaves
(ii) the initial nip
(iii) the initial pre-load required to close the gap 'C' between extra full-length leaves and graduated length leaves.

- VIII. A bracket carrying a load of 15kN is to be welded as shown in figure. Find the size of weld required if the allowable shear stress is not to exceed 80MPa. (15)



OR

- IX. The layout of a transmission shaft carrying two pulleys B and C and supported on bearings A and is shown in figure. Power is supplied to the shaft by means of a vertical belt on the pulley B, which is then transmitted to the pulley C carrying a horizontal belt. The maximum tension in the belt on the pulley B is 2.5kN. The angle of wrap for both the pulleys is 180° and the coefficient of friction is 0.24. The shaft is made of plain carbon steel 30 C₈ (S_{yt} = 400N/mm²) and the factor of safety is 3. Determine the shaft diameter on strength basis. (15)

